

PROJECT TITLE: EV Charging Demand Forecasting

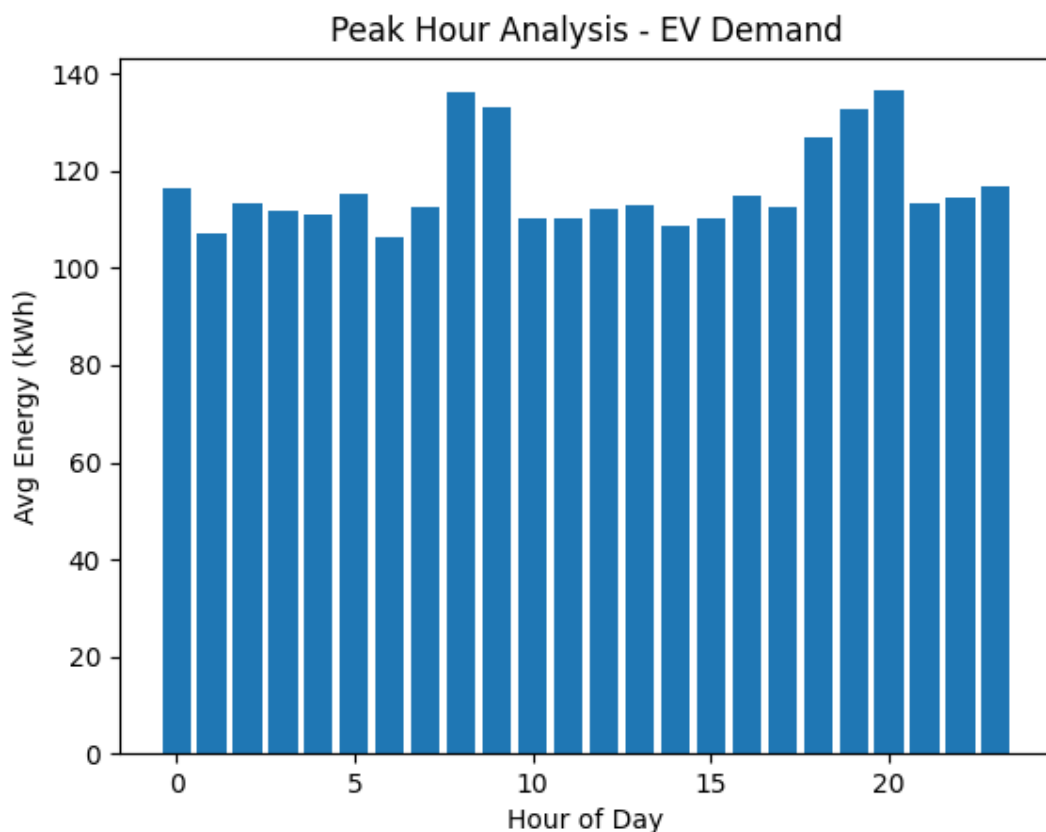
1. Objective:

To predict energy consumption (kWh) at EV charging stations using temperature, traffic flow, and charging price.

2. Dataset Used:

1500 records with columns - timestamp, temperature, traffic_flow, charging_price, hour, station_id, energy_consumed_kWh

3. EDA Insights :



Peak Hours: Demand is highest at 8-9 AM and 6-8 PM (Office hours).] Traffic Impact: Strong positive correlation between traffic_flow and energy demand.

4. Model Used:

Linear Regression

Features: temperature, traffic_flow, charging_price, hour

Target: energy_consumed_kWh

5. Result:

Model Accuracy: 81.70% on test data. Model can predict future demand.

6. Business Recommendation: Company should add more chargers for 8-9 AM and 6-8 PM slot. Increase price slightly during peak hours to manage load. Station ST03 has highest demand.